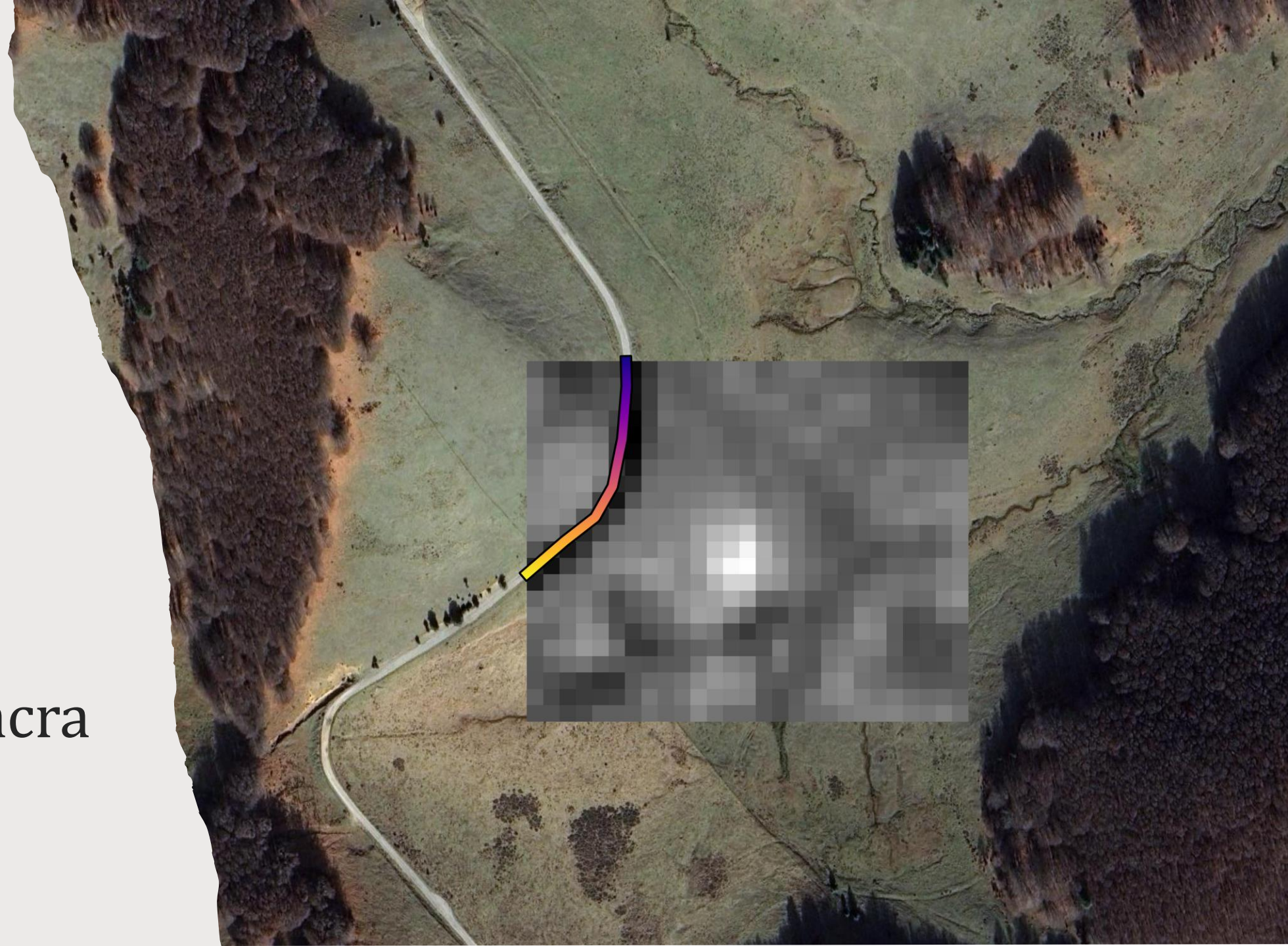


# B3 Hackathon “Post-Print”

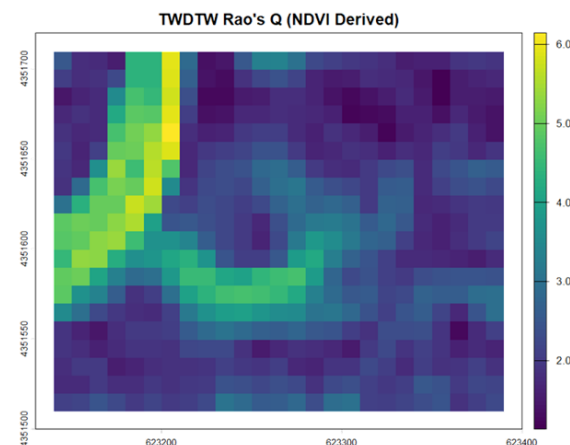
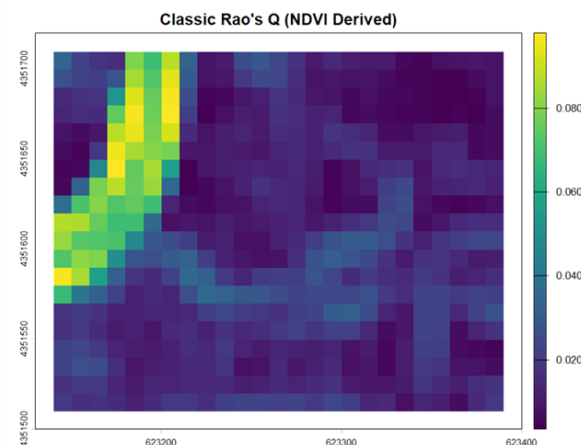
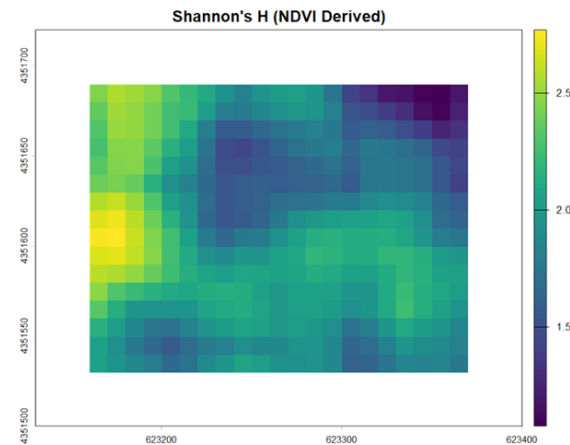
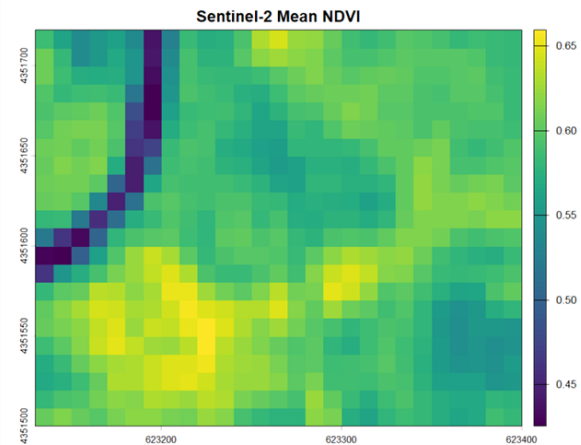
Results as of 23/07/2026

Macchia Sacra



# Macchia Sacra (NDVI-based models)

- PERMANOVAs to compare the classification performance of each index against the ground-truth data
- Model structured as *Diversity Index* ~ *Vegetation Ground Truth*
  - PERMANOVA assessed how much of the spatial variation in each diversity index was explained by vegetation class, using 9,999 permutations.



| <i>Index</i>           | <b>R<sup>2</sup></b> | <b>F</b>  | <b>P value</b> |
|------------------------|----------------------|-----------|----------------|
| <i>Shannon's H</i>     | 0.06019008           | 2.342215  | 0.0241         |
| <i>Classic Rao's Q</i> | 0.08995367           | 3.614908  | 0.0003         |
| <i>TWDTW Rao's Q</i>   | 0.21953164           | 10.286882 | 0.0001         |

# Artifact discrimination

- A Wilcoxon Rank-Sum test to ask “Are the distributions of index values significantly different between road and non-road pixels?”
- The model is:  $Index \sim Road$

```
> MS_NDVI_Wilcox_Shannon
```

```
      wilcoxon rank sum test with continuity correction
```

```
data:  ShannonH by Road
```

```
W = 113, p-value = 0.0000101
```

```
alternative hypothesis: true location shift is not equal to 0
```

```
> MS_NDVI_Wilcox_Classic
```

```
      wilcoxon rank sum test with continuity correction
```

```
data:  RaosQ_Classic by Road
```

```
W = 73, p-value = 0.000005058
```

```
alternative hypothesis: true location shift is not equal to 0
```

```
> MS_NDVI_Wilcox_TWDTW
```

```
      wilcoxon rank sum test with continuity correction
```

```
data:  RaosQ_TWDTW by Road
```

```
W = 151, p-value = 0.00001911
```

```
alternative hypothesis: true location shift is not equal to 0
```

- A Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) test to ask “If I vary every possible threshold of the diversity index, how well can I classify road pixels?”
- The model is:  $P(Road) = f(Index\ value)$

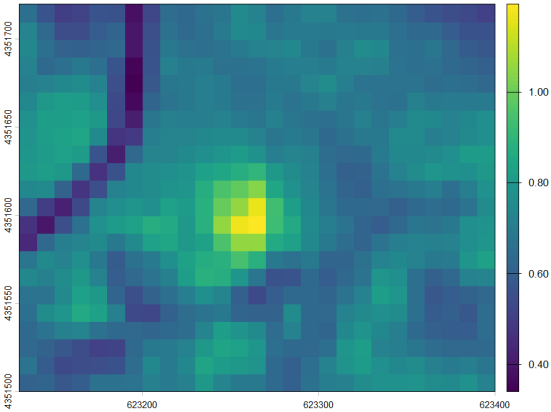
| Index           | Area under curve |
|-----------------|------------------|
| Shannon's H     | 0.9569360        |
| Classic Rao's Q | 0.9721799        |
| TWDTW Rao's Q   | 0.9424543        |



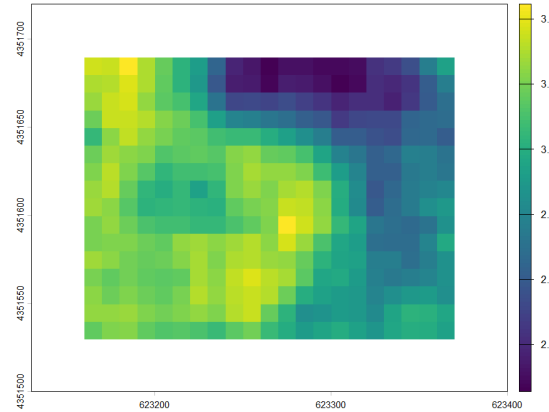
# Macchia Sacra (PPI-based models)

- PERMANOVAs to compare the classification performance of each index against the ground-truth data
- Model structured as *Diversity Index*  $\sim$  *Vegetation Ground Truth*
  - PERMANOVA assessed how much of the spatial variation in each diversity index was explained by vegetation class, using 9,999 permutations.

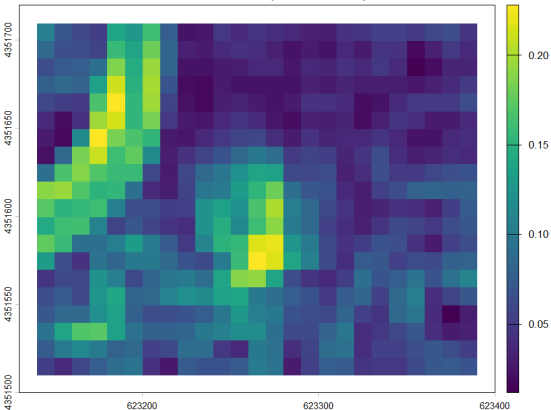
Sentinel-2 Mean PPI



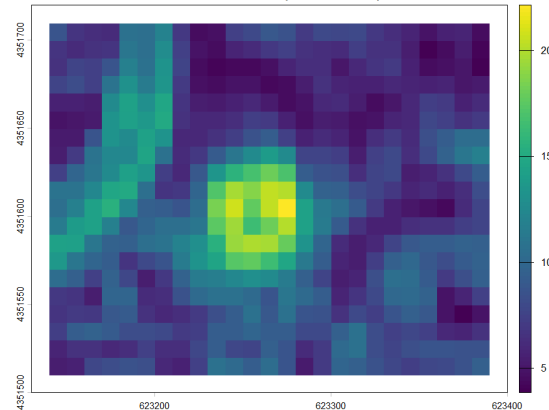
Shannon's H (PPI Derived)



Classic Rao's Q (PPI Derived)



TWDTW Rao's Q (PPI Derived)



| <i>Index</i>           | <b>R<sup>2</sup></b> | <b>F</b> | <b>P value</b> |
|------------------------|----------------------|----------|----------------|
| <i>Shannon's H</i>     | 0.2915129            | 15.04762 | 0.0001         |
| <i>Classic Rao's Q</i> | 0.2538005            | 12.43883 | 0.0001         |
| <i>TWDTW Rao's Q</i>   | 0.4675339            | 32.11169 | 0.0001         |

# Artifact discrimination

- A Wilcoxon Rank-Sum test to ask “Are the distributions of index values significantly different between road and non-road pixels?”
- The model is:  $Index \sim Road$

```
> MS_PPI_wilcox_Shannon
```

```
      wilcoxon rank sum test with continuity correction
```

```
data:  ShannonH by Road
W = 117, p-value = 0.00001081
alternative hypothesis: true location shift is not equal to 0
```

```
> MS_PPI_wilcox_Classic
```

```
      wilcoxon rank sum test with continuity correction
```

```
data:  RaosQ_Classic by Road
W = 84, p-value = 0.00000613
alternative hypothesis: true location shift is not equal to 0
```

```
> MS_PPI_wilcox_TWDTW
```

```
      wilcoxon rank sum test with continuity correction
```

```
data:  RaosQ_TWDTW by Road
W = 435, p-value = 0.001243
alternative hypothesis: true location shift is not equal to 0
```

- A Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) test to ask “If I vary every possible threshold of the diversity index, how well can I classify road pixels?”
- The model is:  $P(Road) = f(Index\ value)$

| Index           | Area under curve |
|-----------------|------------------|
| Shannon's H     | 0.9554116        |
| Classic Rao's Q | 0.9679878        |
| TWDTW Rao's Q   | 0.8342226        |

# Mount Kilimanjaro





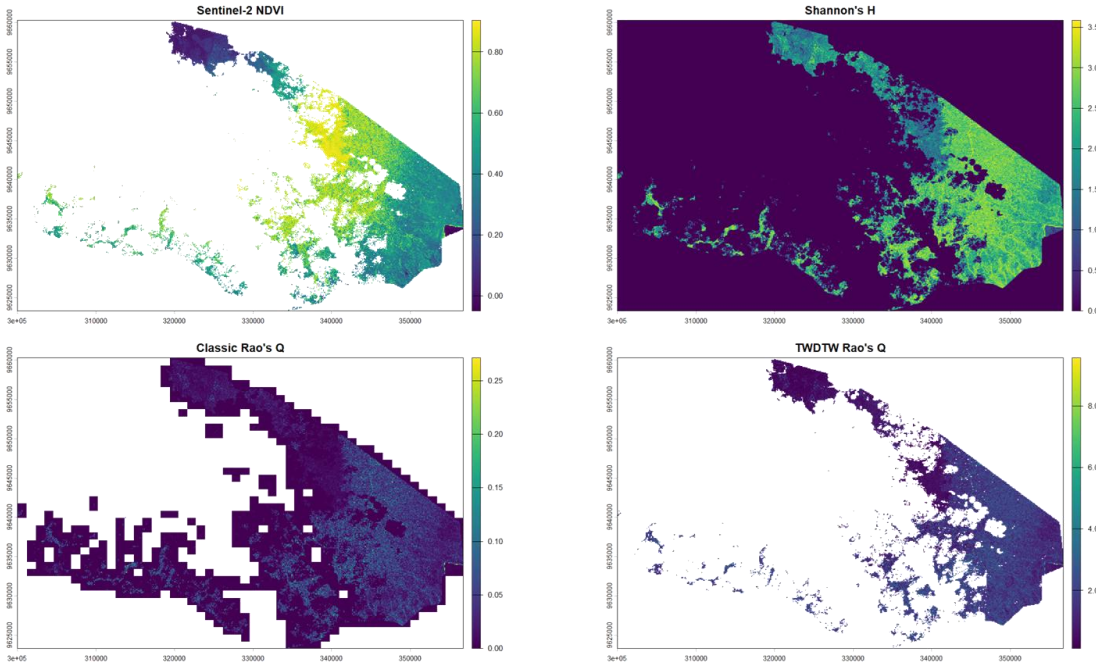
# Kilimanjaro (NDVI-based models)

- PERMANOVAs to compare the classification performance of each index against the ground-truth data
- Model structured as *Diversity Index* ~ *Vegetation Ground Truth*
  - PERMANOVA assessed how much of the spatial variation in each diversity index was explained by vegetation class, using 999 permutations.
- Generated 2 sets of results: using an unmodified NDVI raster, and one using a Savitzky-Golay gap-filled and filtered version of the dataset

Unmodified NDVI raster

| Index | R <sup>2</sup> | F | P value |
|-------|----------------|---|---------|
|-------|----------------|---|---------|

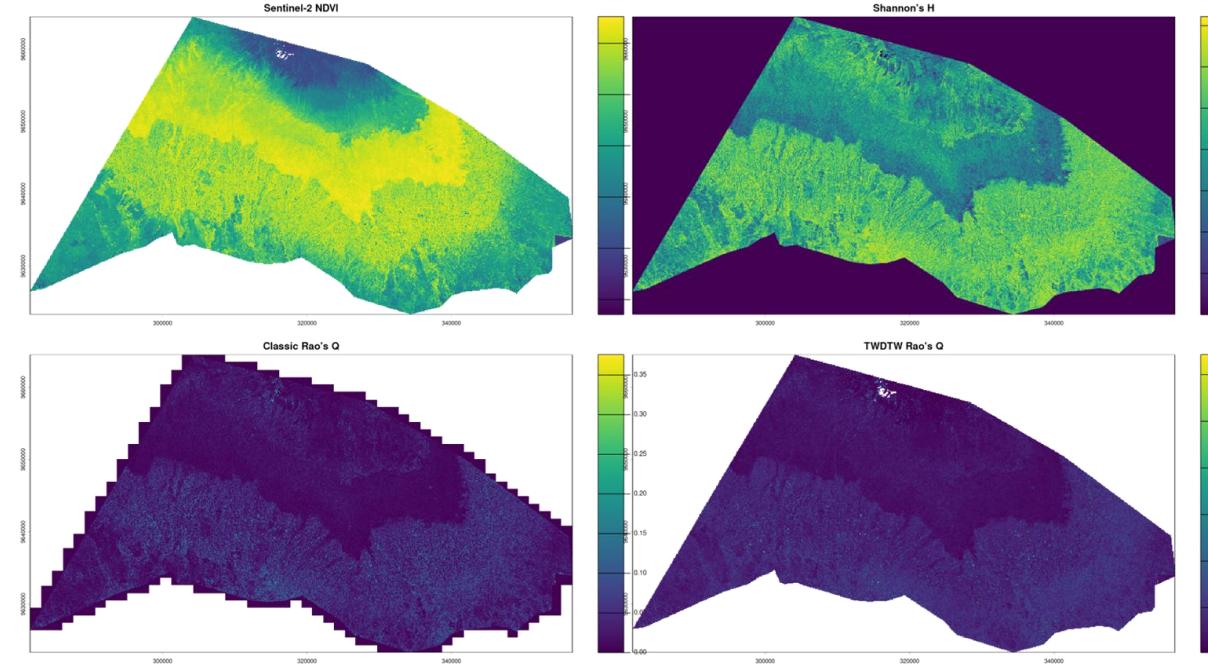
|                 |           |          |       |
|-----------------|-----------|----------|-------|
| Shannon's H     | 0.4540090 | 692.0424 | 0.001 |
| Classic Rao's Q | 0.2621294 | 295.6578 | 0.001 |
| TWDTW Rao's Q   | 0.5285009 | 932.8646 | 0.001 |



Savitzky-Golay gap-filled raster

| Index | R <sup>2</sup> | F | P value |
|-------|----------------|---|---------|
|-------|----------------|---|---------|

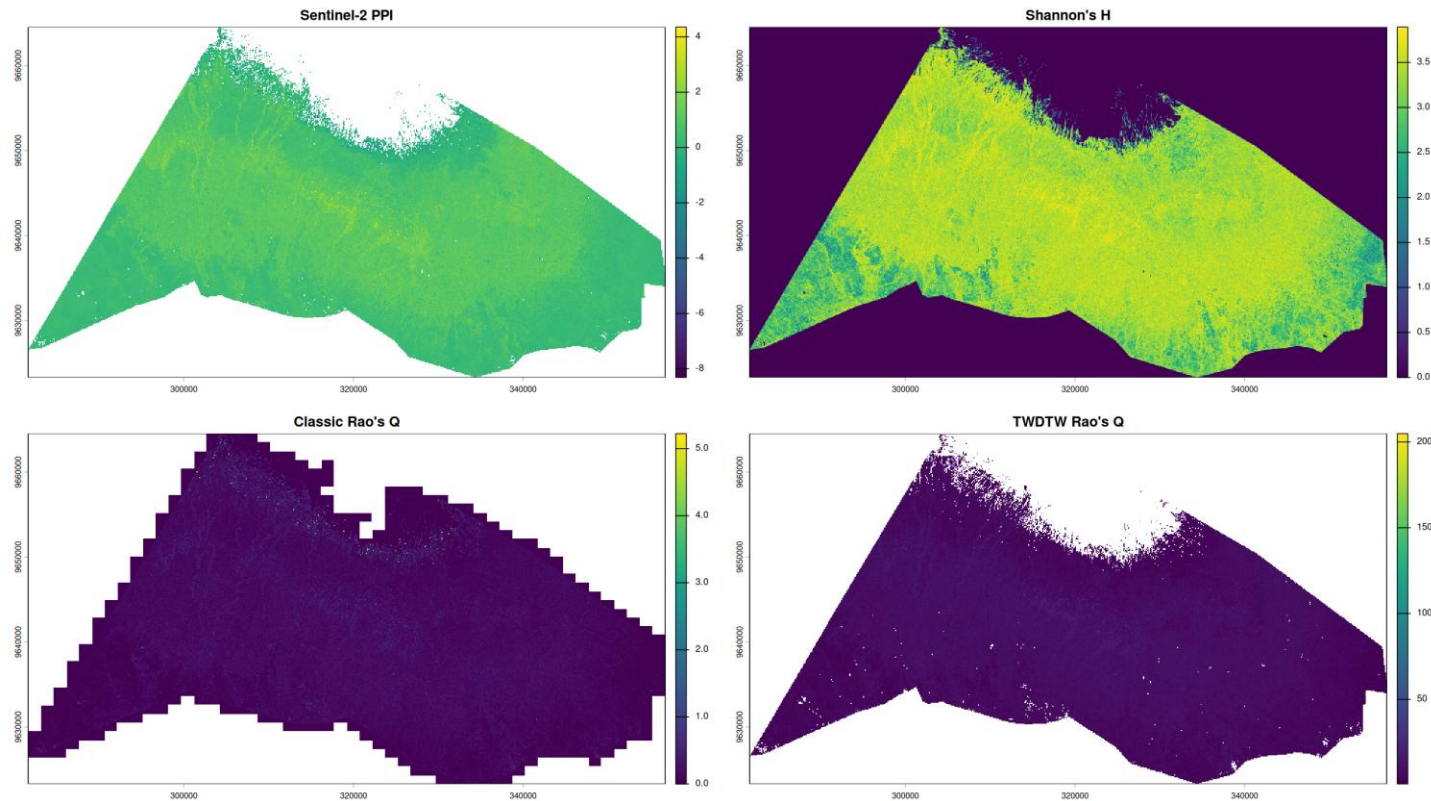
|                 |           |          |       |
|-----------------|-----------|----------|-------|
| Shannon's H     | 0.3881629 | 422.2713 | 0.001 |
| Classic Rao's Q | 0.2254293 | 193.7147 | 0.001 |
| TWDTW Rao's Q   | 0.4273092 | 496.6328 | 0.001 |





# Kilimanjaro (PPI-based models)

- PERMANOVAs to compare the classification performance of each index against the ground-truth data
- Model structured as *Diversity Index* ~ *Vegetation Ground Truth*
  - PERMANOVA assessed how much of the spatial variation in each diversity index was explained by vegetation class, using 999 permutations.

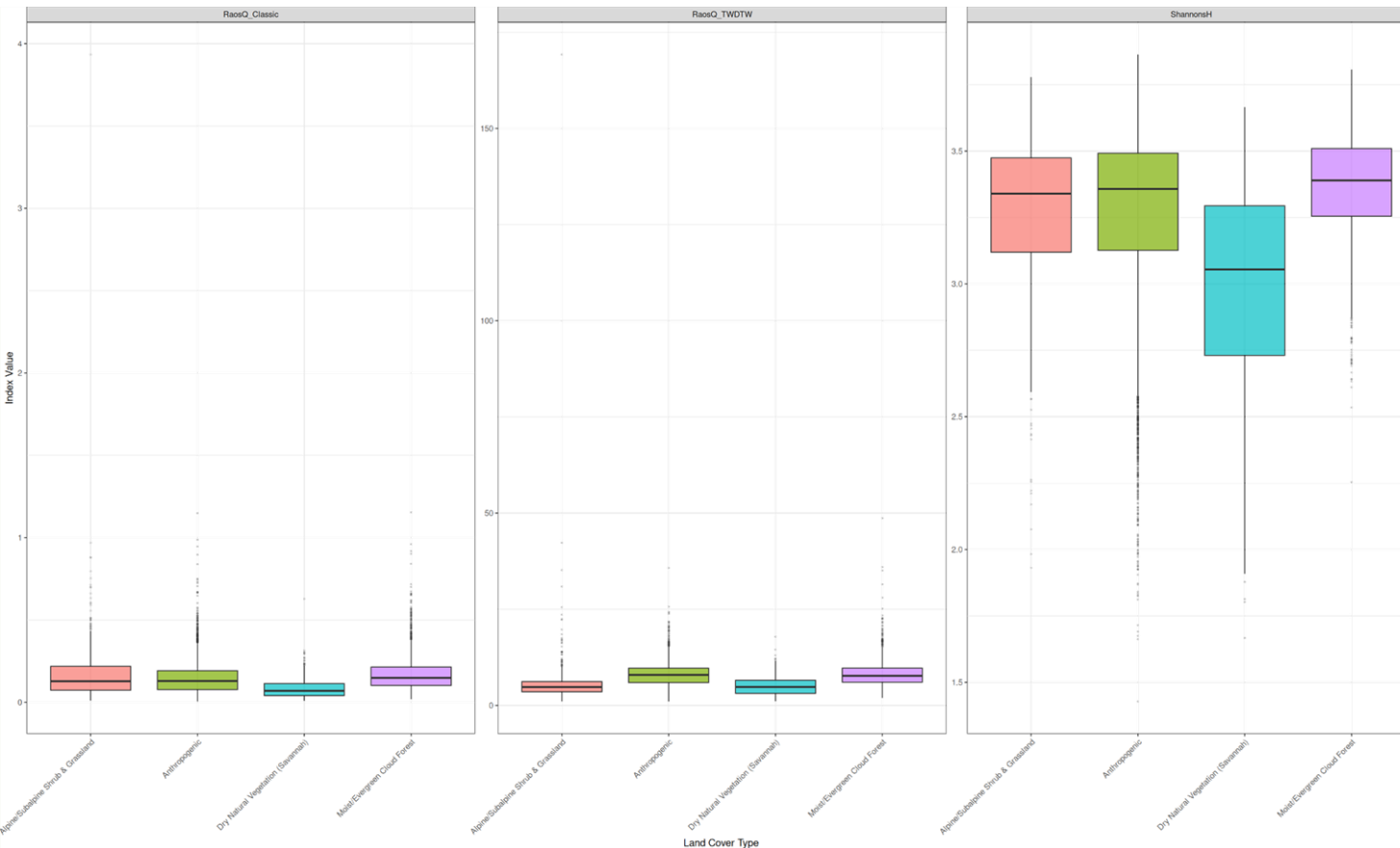


| <i>Index</i>           | <b>R<sup>2</sup></b> | <b>F</b> | <b>P value</b> |
|------------------------|----------------------|----------|----------------|
| <i>Shannon's H</i>     | 0.3222048            | 365.1587 | 0.001          |
| <i>Classic Rao's Q</i> | 0.2006597            | 192.8309 | 0.001          |
| <i>TWDTW Rao's Q</i>   | 0.3147149            | 352.7721 | 0.001          |

# Kilimanjaro (PPI-based models with combined land cover classes)

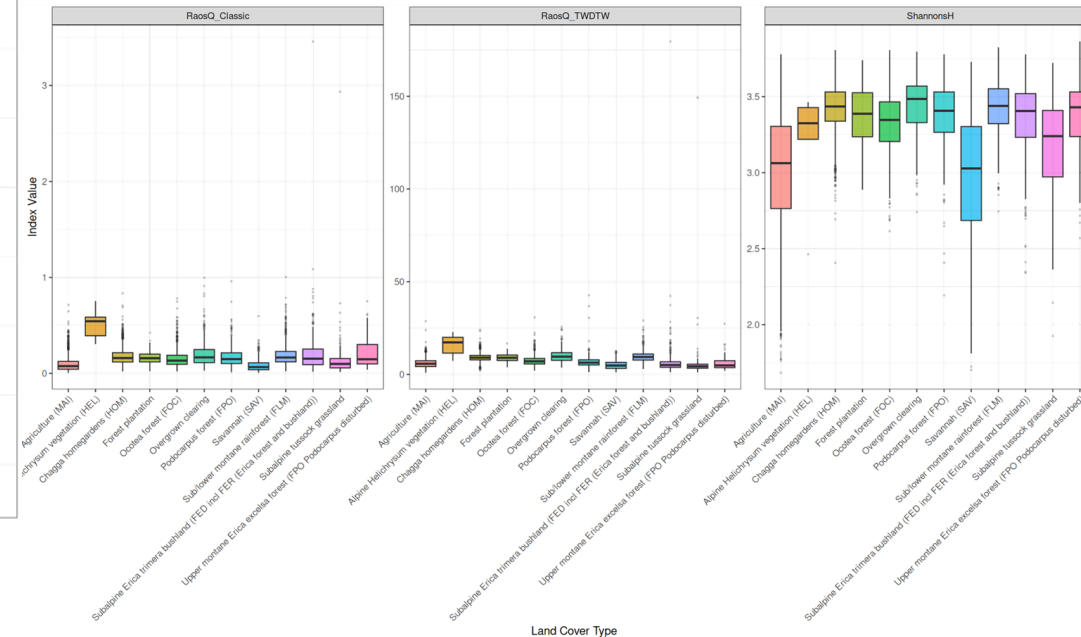
- PERMANOVAs to compare the classification performance of each index against 4 land cover categories which were combined together from the original land cover classes
- Model structured as *Diversity Index ~ Vegetation Ground Truth*
  - PERMANOVA assessed how much of the spatial variation in each diversity index was explained by vegetation class, using 999 permutations.
- I also present box and whisker plots of intra class variance for the original and combined land cover classes

Intra-class Variance: Combined Land Cover Categories



| Index           | R <sup>2</sup> | F        | P value |
|-----------------|----------------|----------|---------|
| Shannon's H     | 0.06249521     | 222.1152 | 0.001   |
| Classic Rao's Q | 0.03627855     | 125.4306 | 0.001   |
| TWDTW Rao's Q   | 0.08687846     | 317.0214 | 0.001   |

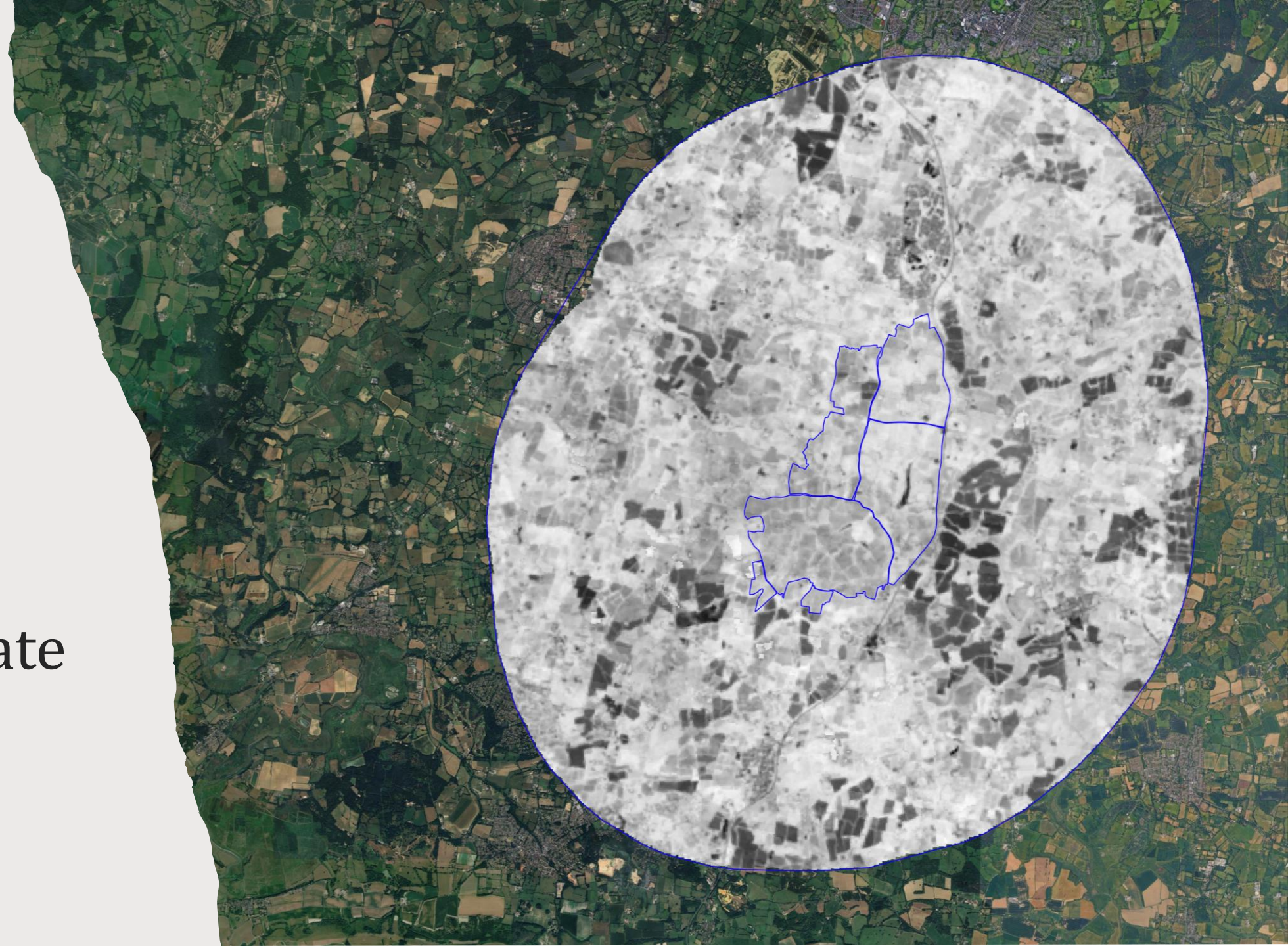
Intra-class Variance: Original Land Cover Categories





# Knepp Estate & environs

Initial results





# Knepp Estate (NDVI-based models from Landsat rasters)

- PERMANOVAs to compare the classification performance of each index against the [land cover classifications from the UK-CEH](#)
  - The UK-CEH's land cover classifications were also derived from Landsat data
- Model structured as *Diversity Index ~ Land Cover Class*
  - PERMANOVA assessed how much of the spatial variation in each diversity index was explained by land cover class, using 999 permutations.
- Results for years 2000, 2007, 2015, and 2020

**Initial results:** I'm awaiting Sentinel-2 rasters, and then I will reassess model performance

|             | 2000           |          |         | 2007           |           |         | 2015           |           |         | 2020           |           |         |
|-------------|----------------|----------|---------|----------------|-----------|---------|----------------|-----------|---------|----------------|-----------|---------|
| Index       | R <sup>2</sup> | F        | p-value | R <sup>2</sup> | F         | p-value | R <sup>2</sup> | F         | p-value | R <sup>2</sup> | F         | p-value |
| Shannon     | 0.08362911     | 65.08879 | 0.001   | 0.1149796      | 117.96505 | 0.001   | 0.13285091     | 170.05669 | 0.001   | 0.14443206     | 240.97029 | 0.001   |
| Classic Rao | 0.0553563      | 41.79449 | 0.001   | 0.08234763     | 81.48146  | 0.001   | 0.0867102      | 105.38639 | 0.001   | 0.11610027     | 187.49282 | 0.001   |
| TWDTW Rao   | 0.05222066     | 39.29662 | 0.001   | 0.0824864      | 81.63111  | 0.001   | 0.08928945     | 108.82853 | 0.001   | 0.0833289      | 129.7587  | 0.001   |